AOU
الجامعة العربية المفتوحة
Arab Open University
Faculty of Computer Studies

T103/TM103
Computer Organization and Architecture

Final Examination
Summer Semester 2025/2026

Time Allowed: 2 Hours

AOU EGYPT

Instructions:

- Total Marks: **50**
- This exam consists of **Three** parts.
- **ALL questions** must be answered in the External Answer booklet.
- Be sure you write your **name and ID** on the External Answer booklet.
- **Calculators** are **NOT** allowed.

الجامعة العربية المفتوحة - مصر
Arab Open University
القبول والتسجيل
الامتحانات
Arab Republic of Egypt

CamScanner

Part I: Choose the most correct answer (10 marks)
This part consists of **10 questions** carrying a **weight of one mark each**.
Q-1: Answer all the following questions by choosing the most correct statement.

1. If a carry-out of value 1 from the 8th bit occurs when adding two signed numbers using One's Complement representation, then:
   a. The result cannot be known.
   b. We have an erroneous result.
   c. It will be added to the primary result.
   d. It will be discarded.
2. Which of the following is a sequential circuit?
   a. Full Adder
   b. Register
   c. Multiplexer
   d. Both a and c
3. If a function F is equal to **0** for the expression **(a⊕b)'**, then:
   a. All inputs are 1, exclusively
   b. All inputs are 0, exclusively
   c. Only one input is 1
   d. All inputs are equal
4. Which of the following is true about Dynamic RAM?
   a. Electricity is always applied.
   b. Cheaper than SRAM
   c. Used as a Cache memory.
   d. Faster than SRAM.
5. Which of the following is a valid statement?
   a. The MARIE Instruction Set is composed of 8 instructions.
   b. A miss is recorded when the data is found at a given memory level.
   c. K-Maps are used to simplify a logical expression.
   d. None of the choices.
6. Based on what you have studied in this course, the smallest possible *addressable* unit of computer storage is the:
   a. Bit
   b. Byte
   c. Word
   d. Register
7. Which of the following is a valid statement?
   a. Full adders have 3 inputs
   b. RAM is faster than Cache
   c. Registers are in the CPU.
   d. Both a and c
8. If we want to build a **32Kx16** word addressable RAM memory using only **8Kx8 RAM** chips, then we need:
   a. 32 **8Kx8** RAM chips
   b. 64 **8Kx8** RAM chips
   c. 8 **8Kx8** RAM chips
   d. 16 **8Kx8** RAM chips
9. The number of registers in MARIE is:
   a. 13.
   b. 16.
   c. 6.
   d. 7.
10. The flip-flop that has an unstable (invalid) state is called:
    a. J.K
    b. S.R
    c. D
    d. Either a or b

الجامعة العربية المفتوحة - مصر
Arab Open University
القبول والتسجيل
الامتحانات
Arab Republic of Egypt

CamScanner

Part II: Short essay questions (15 marks)

This part consists of **five questions** carrying a **WEIGHT OF 3 marks each**.
You should answer **all** the questions.

Q-2: What do the following acronyms stand for: EPROM, CPU, MBR?

Q-3: Suppose that we have a **128M * 64**-bit memory module (**Byte addressable**):

a. What is the word size?
b. How many bits are needed to represent each address in this module? Show your calculations.

Q-4: Draw the truth table of the Half-adder and its Logical Diagram.

Q-5: Using MARIE instruction set, find the equivalent assembly codes of the following binary instructions. N.B: The MARIE instruction has the following format:



a. **0100** 1110 **0111** 0000.
b. **1001** 1010 **0110** 0000.

Q-6: Referring to MARIE ISA, what do **Load X** and **Input** do?

Part III: Numerical problems (25 marks)

This part consists of **four questions** carrying a weight of **25 marks**.
You should **answer all the questions**.

Q-7: Suppose that you have the following pseudocode excerpt: **[6 marks]**

```
Input a number
Assign this number
to x
Input a number
Assign this number
to y
Let z=y-x
Jump to the End
Label
Let AC= x
End: Display the
AC value
Stop
```

a) Write a MARIE assembly code to perform the above pseudocode excerpt.
b) What will be the output if the entered numbers are 5 and 9 respectively?



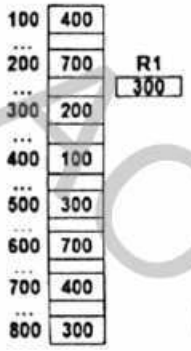
AOU EGYPT

CamScanner

Q-8: Suppose we have the following MARIE code: **[6 marks]**

Load 200
Add 100

Knowing that **AC=100** before executing the two instruction, and the memory and register R1 containing the values below, determine the actual value that will be found in the accumulator **AC** after executing each instruction in each mode, and fill in the table below: Note that for each addressing mode the value of AC is initially 100



Mode	Value in AC after Load 200	Value in AC after Add 100
Immediate		
Direct		
Indirect		
Indexed		

Q-9: Answer the following questions: **[6 marks]**

a) Find the values of (-68) and (-64) in 8-bits binary representation using **C2** representation.
b) Find the sum of (-68) and (-64) in 8-bits binary representation using **C2** representation.
c) What can you say about the result? Discuss.

Q-10: Assume that we have the following Direct Mapped Cache representation:

Direct Mapped

3	15	6

Answer the following questions based on the above data: **[7 marks]**

a- What is the size of the main memory?
b- What is the number of blocks in the cache memory?
c- How many blocks are there in the main memory?

Draw the format of the memory address as seen by the cache, showing the fields as well as their sizes, for the following scheme:

d- 8-way set associative cache mapping scheme

End of questions

الجامعة العربية المفتوحة - مصر
Arab Open University
القبول والتسجيل
الامتحانات
Arab Republic of Egypt

AOU EGYPT
CamScanner